
MODEL SELECTION AND VALIDATION REPORT (2017_2022_biomass model)

OBSERVATIONS:945

SANITY: OK - the model passed sanity check.

MANUALLY SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (2017_2022_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Mesh: mesh (isotropic covariance)

Time column: year

Data: data

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.22	0.17
kde_fric	0.17	0.03
kde_feve	-0.19	0.03
tric	-0.05	0.03
teve	-0.44	0.03
sbt_mean	0.36	0.08
sbt_sd	-0.02	0.05
sst_sd	0.01	0.05
log_chla_mean	0.08	0.04
depth	-0.13	0.04

Dispersion parameter: 0.69

Spatiotemporal AR1 correlation (rho): 0.81

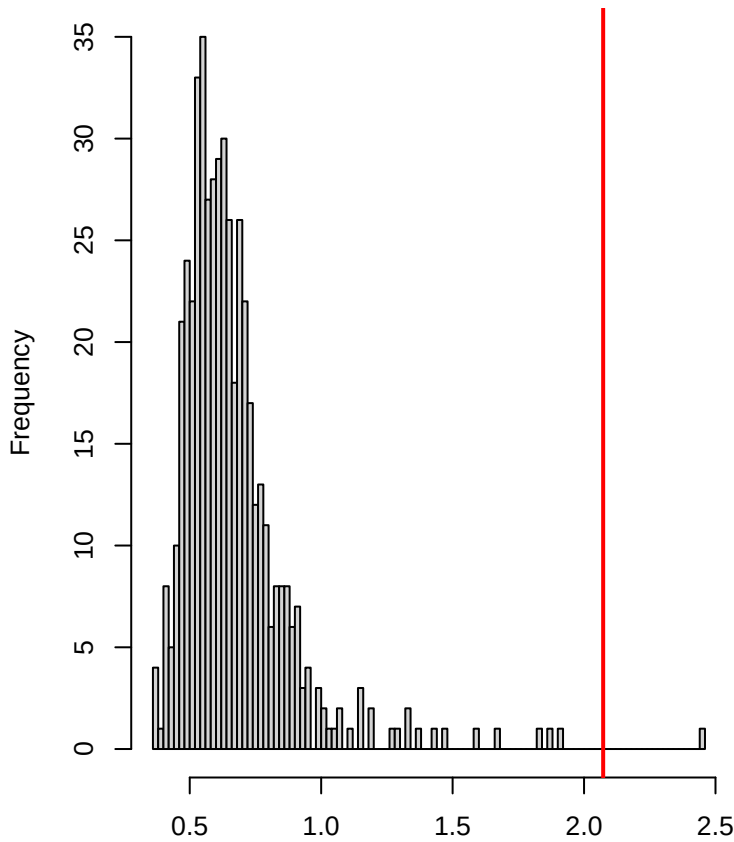
Matérn range: 347.21

Spatiotemporal marginal AR1 SD: 0.53

REML criterion at convergence: 7689.097

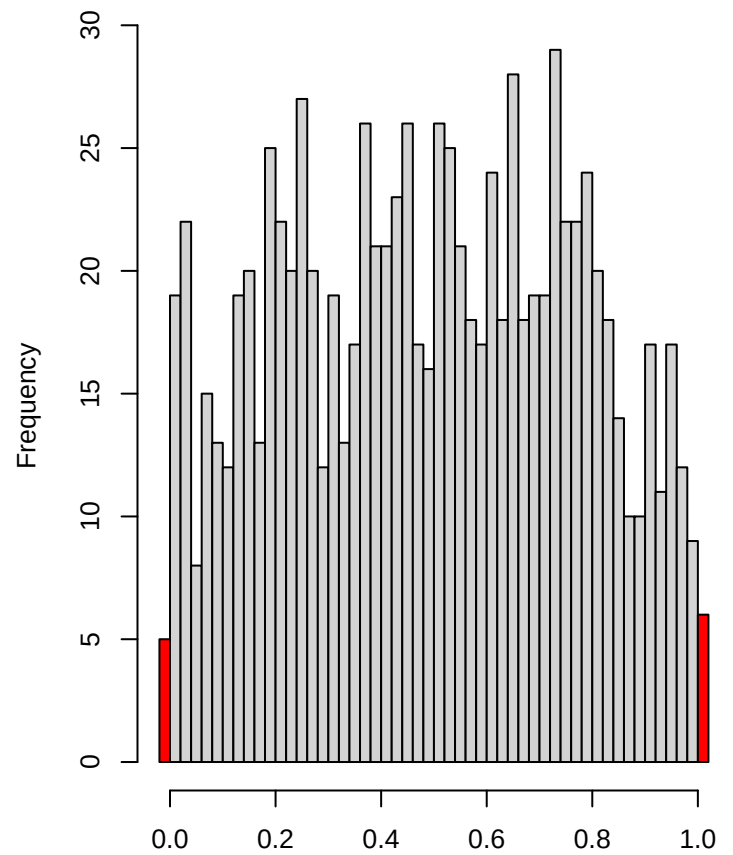
See ?tidy.sdmTMB to extract these values as a data frame.

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



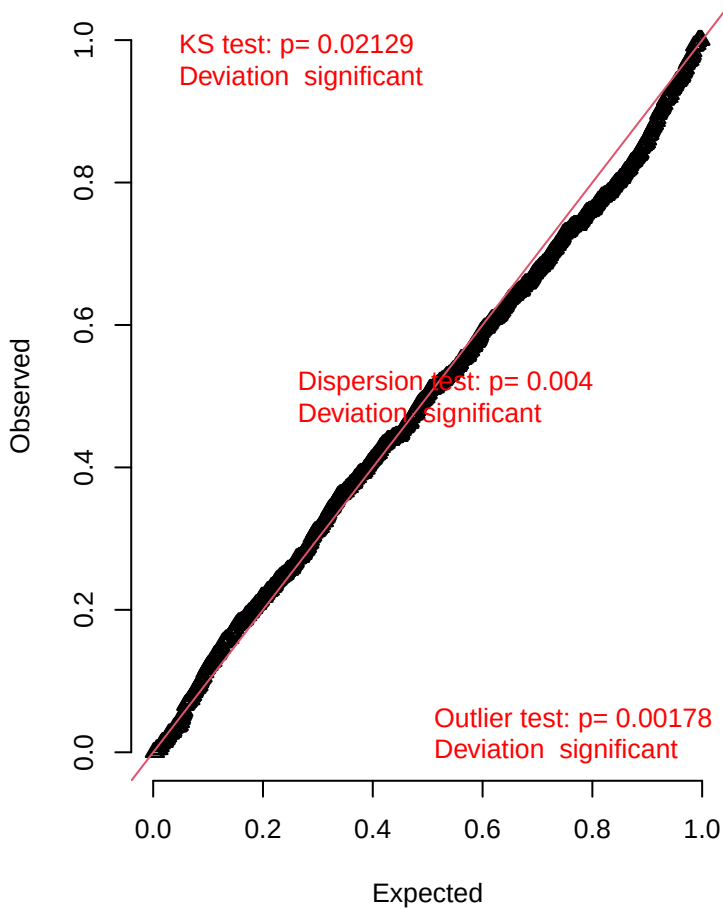
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

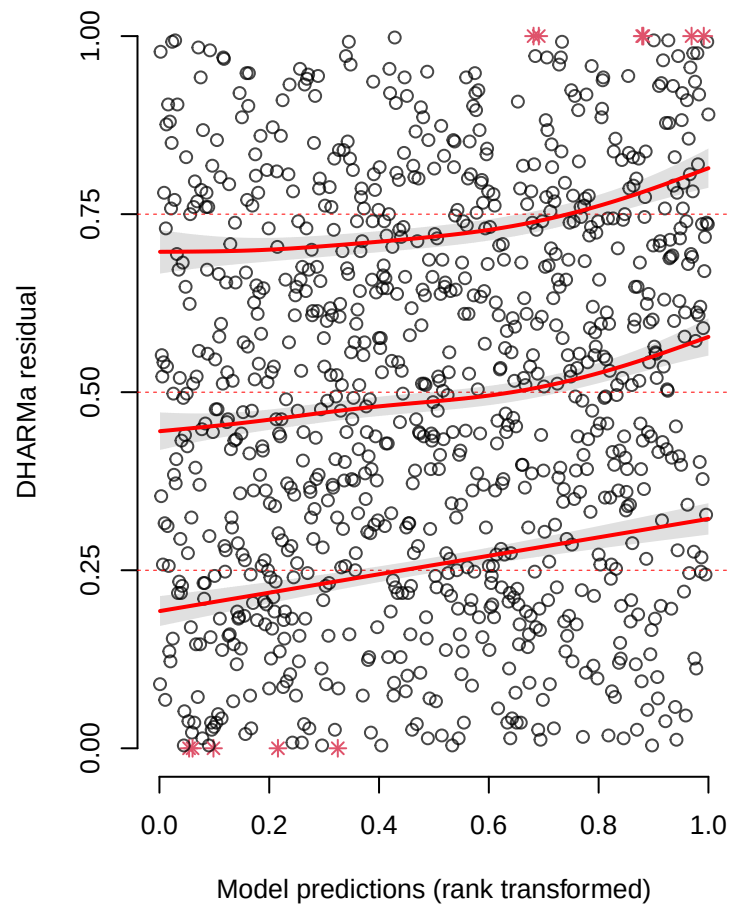


Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.018 | H1: Distance-based autocorrelation | p-value = 0.271

